

MODULE – 2

LINEAR ALGEBRA

A matrix in mathematics is a rectangular array or grid of numbers, symbols, or expressions arranged in rows and columns, typically enclosed in brackets.

Representation:

$$A_{ij} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$$

Operations on Matrices

Matrix operations are basic concepts in linear algebra that are widely used in mathematics, engineering, and data science. The commonly used matrix operations are **addition, subtraction, scalar multiplication, matrix multiplication, transpose, and inverse**. The rules for each operation are explained below in detail.

Matrix Addition

Two matrices can be added only if they have the same order (same number of rows and columns).

Rule: If $A = [a_{ij}]$ and $B = [b_{ij}]$ of order $m \times n$ then $A + B = [a_{ij} + b_{ij}]$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$
$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

Properties

- Commutative: $A + B = B + A$
- Associative: $(A + B) + C = A + (B + C)$
- Additive Identity: $A + 0 = A$

Matrix Subtraction

Matrix subtraction is possible only when matrices are of the same order.

Rule: If $A = [a_{ij}]$ and $B = [b_{ij}]$ of order $m \times n$ then $A - B = [a_{ij} - b_{ij}]$

Example:

$$A = \begin{bmatrix} 8 & 6 \\ 4 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 3 \\ 2 & 1 \end{bmatrix}$$
$$A - B = \begin{bmatrix} 8-1 & 6-3 \\ 4-2 & 2-1 \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 2 & 1 \end{bmatrix}$$

Scalar Multiplication

When a matrix is multiplied by a scalar (a real number), each element of the matrix is multiplied by that scalar.

Rule: If k is a scalar and $A = [a_{ij}]$ then $kA = [k \cdot a_{ij}]$

Example:

$$A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} \quad k = 3$$
$$3A = 3 \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$
$$= \begin{bmatrix} 3 \times 2 & 3 \times 4 \\ 3 \times 6 & 3 \times 8 \end{bmatrix}$$
$$= \begin{bmatrix} 6 & 12 \\ 18 & 24 \end{bmatrix}$$

Matrix Multiplication

Matrix multiplication is defined only when the number of columns of the first matrix is equal to the number of rows of the second matrix.

Rule: If A is of order $m \times n$ and B is of order $n \times p$, then AB is the order of $m \times p$. Each element of AB is obtained by multiplying the corresponding row of A with the column of B .

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} (1 \times 5 + 2 \times 7) & (1 \times 6 + 2 \times 8) \\ (3 \times 5 + 4 \times 7) & (3 \times 6 + 3 \times 8) \end{bmatrix}$$

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Properties

- Not commutative: $AB \neq BA$
- Associative: $(AB)C = A(BC)$
- Distributive: $A(B + C) = AB + AC$

Transpose of a Matrix

The transpose of a matrix is obtained by interchanging rows and columns.

Rule: If $A = [a_{ij}]$ then transpose $A^T = [a_{ji}]$

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Properties:

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$

Inverse of a matrix

The inverse of a matrix exists only for square matrices and only when the determinant is non- zero.

Rule: If A is a square matrix and $|A| \neq 0$ [$\det(A)$], then

$$AA^{-1} = A^{-1}A = I$$

Example:

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

$$\begin{aligned}|A| &= \begin{vmatrix} 2 & 1 \\ 5 & 3 \end{vmatrix} = (2 \times 3) + (5 \times 1) \\ &= 6 - 5 \\ &= 1\end{aligned}$$

Since $|A| \neq 0$ inverse exists

$$A^{-1} = \frac{1}{|A|} (\text{Adjoint } (A))$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$$

RANK AND NULLITY OF A MATRIX

Rank of a Matrix

The rank of a matrix is the maximum number of linearly independent rows or columns in the matrix. (OR) Total number of non-zero rows in the matrix.

Null Space

The null space of a matrix A is the set of all vectors X such that $AX = 0$

It represents solutions of a homogeneous system of equations.

Nullity

The nullity of a matrix is the dimension of its null space, i.e., the number of free variables.

Important Relation (Rank–Nullity Theorem)

Rank + Nullity = Number of columns of the matrix

Find the rank of the matrix

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 5 \\ 3 & 7 & 8 \end{bmatrix}$$

Sol:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 5 \\ 3 & 7 & 8 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 3R_1$$

$$A \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -5 & 1 \\ 0 & -2 & 2 \end{bmatrix}$$

$$R_3 \rightarrow 5R_3 - 2R_2$$

$$A \sim \begin{bmatrix} 1 & 3 & 2 \\ 0 & -5 & 1 \\ 0 & 0 & 8 \end{bmatrix}$$

$$\rho(A) = 3$$

$\therefore$ Rank of the matrix A is 3

ROW 1		ROW 2
$R_2 \rightarrow R_2 - 2R_1$	$R_3 \rightarrow R_3 - 3R_1$	$R_3 \rightarrow 5R_3 - 2R_2$
$2 - 2(1) = 0$	$3 - 3(1) = 0$	0
$1 - 3(2)$ $= 1 - 6$ $= -5$	$7 - 3(3)$ $7 - 9$ $= -2$	$5(-2) - 2(-5)$ $= -10 + 10$ $= 0$
$5 - 2(2)$ $= 5 - 4$ $= 1$	$8 - 3(2)$ $= 8 - 6$ $= 2$	$5(2) - 2(1)$ $= 10 - 2$ $= 8$

PSEUDO INVERSE OF A MATRIX

The concept of Pseudo Inverse is also called **Moore – Penrose Pseudo inverse** which is used to compute the best fit (least squares) solution to a system of linear solution that lacks unique solution.

It is denoted by the symbol A^+

To compute the pseudo inverse of a matrix, we must know the concept of

- a. Full Row Rank
- b. Full Column Rank
- c. Rank Deficient System

Full Row Rank: Rank of matrix of order $m \times n$ is equal to the number of rows in the matrix.

Full Column Rank: Rank of matrix of order $m \times n$ is equal to the number of columns in the matrix.

Rank Deficient System: Rank of matrix of order $m \times n$ is less than the number of rows or columns in the matrix.

For these, the governing formulas are

Full Row Rank: If the rows are linearly independent, pseudo inverse is given as

$$A^+ = (A)^T \cdot (AA^T)^{-1}$$

Full Column Rank: If the columns are linearly independent, pseudo inverse is given as

$$A^+ = (A^T A)^{-1} \cdot (A)^T$$

If the system of equations is Rank Deficient

$$A^+(U\Sigma V^T)^{-1} = (V^T)^{-1}\Sigma^{-1}U^{-1} = V\Sigma^{-1}U^{-1}$$

$$\text{If } \Sigma = \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \\ 0 & 0 \end{bmatrix} \quad \Sigma^{-1} = \begin{bmatrix} 1/\sigma_1 & 0 \\ 0 & 1/\sigma_2 \\ 0 & 0 \end{bmatrix}$$

Problems

i. Find Pseudo inverse of

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

Sol: Given

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{bmatrix}$$

$$\text{Here } \begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix} = 3 - 8 = -5 \neq 0$$

$$\rho(A) = 2$$

Since rank = Number of Rows, the problem must be solved using row rank

$$A^+ = (A)^T \cdot (AA^T)^{-1}$$

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 4 \\ 2 & 3 \\ 1 & 2 \\ 3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{Adj}A$$

$$AA^T = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 4 & 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \\ 1 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 15 & 15 \\ 15 & 30 \end{bmatrix}$$

$$|AA^T| = 15(30 - 15) = 225$$

$$(AA^T)^{-1} = \frac{1}{225} \begin{vmatrix} 30 & -15 \\ -15 & 15 \end{vmatrix} = \begin{bmatrix} 2/15 & -1/15 \\ -1/15 & 1/15 \end{bmatrix}$$

$$A^+ = (A)^T \cdot (AA^T)^{-1}$$

$$= \begin{bmatrix} 1 & 4 \\ 2 & 3 \\ 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2/15 & -1/15 \\ -1/15 & 1/15 \end{bmatrix}$$

$$= \begin{bmatrix} -2/15 & 3/15 \\ 1/15 & 1/15 \\ 0/15 & 1/15 \\ 5/15 & -2/15 \end{bmatrix}$$

$$= \frac{1}{15} \begin{bmatrix} -2 & 3 \\ 1 & 1 \\ 0 & 1 \\ 5 & -2 \end{bmatrix}$$

ii. Find the Pseudo Inverse of the matrix for

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -1 \end{bmatrix}$$

Sol: Given

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -1 \end{bmatrix}$$

Here $\begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 - 4 = -3 \neq 0$

$$\rho(A) = 2$$

The given problem follows column rank

$$A^+ = (A^T A)^{-1} \cdot (A)^T$$

$$A^T A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix}$$

$$|A^T A| = \begin{vmatrix} 6 & 3 \\ 3 & 6 \end{vmatrix} = 36 - 9 = 27$$

$$(A^T A)^{-1} = \frac{1}{27} \begin{vmatrix} 6 & -3 \\ -3 & 6 \end{vmatrix} = \begin{bmatrix} 2/9 & -1/9 \\ -1/9 & 2/9 \end{bmatrix}$$

$$A^+ = (A^T A)^{-1} \cdot (A)^T = \begin{bmatrix} 2/9 & -1/9 \\ -1/9 & 2/9 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 3/9 & 3/9 \\ 3/9 & 0 & -3/9 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

EIGEN VALUE DECOMPOSITION

Eigen decomposition is a method used in linear algebra to break down a square matrix into simpler components called eigenvalues and eigenvectors. This process helps us understand how a matrix behaves and how it transforms data.

We represent the Eigen Value Decomposition using the formula

$$A = PDP^{-1}$$

But in the case of symmetric matrices, we express the same as

$$A = PDP^T$$

Problems

i. Find the Eigen Decomposition of the matrix

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Sol: The characteristic equation

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{vmatrix} = 0$$

$$(4 - \lambda)(3 - \lambda) - 2 \times 1 = 0$$

$$12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$\lambda^2 - 5\lambda - 2\lambda + 10 = 0$$

$$\lambda(\lambda - 5) - 2(\lambda - 5) = 0$$

$$(\lambda - 5)(\lambda - 2) = 0$$

$$\lambda = 5 \quad \lambda = 2$$

Case I: When $\lambda = 5$

$$\begin{bmatrix} 4-5 & 1 \\ 2 & 3-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$-1x + 1y = 0$$

$$2x - 2y = 0$$

From i

$$y = x$$

$$\Rightarrow X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Case I: When $\lambda = 2$

$$\begin{bmatrix} 4-2 & 1 \\ 2 & 3-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$2x + y = 0$$

$$2x + y = 0$$

From i

$$2x = -y$$

$$x = -1/2$$

When substituted back to $2x = -y$

$$2\left(-\frac{1}{2}\right) = -y$$

$$y = 1$$

$$\Rightarrow X_2 = \begin{bmatrix} -1/2 \\ 1 \end{bmatrix}$$

Now

$$A = PDP^{-1}$$

$$P = \begin{bmatrix} 1 & -1/2 \\ 1 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

$$P^{-1} = \frac{1}{|P|} \text{Adj}P$$

$$|P| = (1 \times 1) - \left(\frac{-1}{2} \times 1 \right)$$

$$= 1 + \frac{1}{2} = \frac{3}{2}$$

$$P^{-1} = \frac{1}{3/2} \begin{vmatrix} 1 & -1/2 \\ -1 & 1 \end{vmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ -2/3 & 2/3 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -1/2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 \\ -2/3 & 2/3 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

ii. Find Eigen Value Decomposition for the matrix

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

$$\lambda I = \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol: The characteristic equation is given by the formula $|A - \lambda I| = 0$

$$\begin{vmatrix} 8 - \lambda & -6 & 2 \\ -6 & 7 - \lambda & -4 \\ 2 & -4 & 3 - \lambda \end{vmatrix} = 0$$

$$= \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\lambda^3 - (\sum d)\lambda^2 - (\sum m_d)\lambda - |A| = 0$$

$$\sum d = 8 + 7 + 3 = 18 \text{ (Sum of all diagonal elements)}$$

$$\begin{aligned}\sum m_d &= \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} \\ &= (21 - 16) + (24 - 4) + (56 - 36)\end{aligned}$$

$$= 45 \text{ (Summation other than diagonal elements)}$$

$$\begin{aligned}|A| &= \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix} = 8(21 - 16) + 6(-18 + 8) + 2(24 - 14) \\ &= 8(5) + 6(-10) + 2(10) \\ &= 40 - 60 + 20 = 60 - 60 = 0\end{aligned}$$

The equation can be written as

$$\lambda^3 - 18\lambda^2 + 45\lambda - 0 = 0$$

$$\lambda^3 - 18\lambda^2 + 45\lambda = 0$$

$$\lambda(\lambda^2 - 18\lambda + 45) = 0$$

$$\lambda = 0 \quad \text{or} \quad \lambda^2 - 18\lambda + 45 = 0$$

$$\lambda^2 - 15\lambda - 3\lambda + 45 = 0$$

$$\lambda(\lambda - 15) - 3(\lambda - 15) = 0$$

$$(\lambda - 15)(\lambda - 3) = 0$$

$$\Rightarrow \lambda = 15 \Rightarrow \lambda = 3$$

$\therefore$ The eigen values are $\lambda_1 = 0, \lambda_2 = 3, \lambda_3 = 15$

Now we form the system of equations

$$[A - \lambda I][X] = 0$$

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$(8-\lambda)x - 6y + 2z = 0$$

$$-6x + (7-\lambda)y - 4z = 0$$

$$2x - 4y + (3-\lambda)z = 0$$

Case 1: When $\lambda = 0$ eq 1 becomes

$$8x - 6y + 2z = 0 \longrightarrow \text{i}$$

$$-6x + 7y - 4z = 0 \longrightarrow \text{ii}$$

$$2x - 4y + 3z = 0 \longrightarrow \text{iii}$$

Applying Cross Multiplication Rule on i and ii

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 7 & -4 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 8 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{-z}{\begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}} \begin{matrix} \text{ii} \\ \text{iii} \end{matrix}$$

$$\Rightarrow \frac{x}{24 - 14} = -\frac{y}{-32 + 12} = \frac{z}{56 - 36}$$

$$\Rightarrow \frac{x}{10} = \frac{-y}{-20} = \frac{z}{20} \times 10$$

$$\Rightarrow \frac{x}{1} = \frac{y}{2} = \frac{z}{2}$$

At Eigen value $\lambda_1 = 0$, the eigen vector $X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

Case 2: When $\lambda = 3$ eq 1 becomes

$$5x - 6y + 2z = 0 \longrightarrow \text{i}$$

$$-6x + 4y - 4z = 0 \longrightarrow \text{ii}$$

$$2x - 4y + 0z = 0$$

Applying Cross Multiplication Rule on i and ii

$$\begin{aligned}\frac{x}{\begin{vmatrix} -6 & 2 \\ 4 & -4 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} 5 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 5 & -6 \\ -6 & 7 \end{vmatrix}} \\ \Rightarrow \frac{x}{24 - 8} &= -\frac{y}{-20 + 12} = \frac{z}{20 - 36} \\ \Rightarrow \frac{x}{16} &= \frac{-y}{-8} = \frac{z}{-16} \times 8 \\ \Rightarrow \frac{x}{2} &= \frac{y}{1} = \frac{z}{-2}\end{aligned}$$

At Eigen value $\lambda_2 = 3$, the eigen vector $X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$

Case 2: When $\lambda = 15$ eq 1 becomes

$$-7x - 6y + 2z = 0 \longrightarrow \text{ i }$$

$$-6x - 8y - 4z = 0 \longrightarrow \text{ ii }$$

$$2x - 4y - 12z = 0 \longrightarrow \text{ iii }$$

Applying Cross Multiplication Rule on i and ii

$$\begin{aligned}\frac{x}{\begin{vmatrix} -6 & 2 \\ -8 & -4 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} -7 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -7 & -6 \\ -6 & -8 \end{vmatrix}} \\ \Rightarrow \frac{x}{24 + 16} &= -\frac{y}{28 + 12} = \frac{z}{56 - 36} \\ \Rightarrow \frac{x}{40} &= \frac{-y}{40} = \frac{z}{20} \times 20 \\ \Rightarrow \frac{x}{2} &= \frac{y}{-2} = \frac{z}{1}\end{aligned}$$

At Eigen value $\lambda_3 = 15$, the eigen vector $X_2 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

The eigen values are $\lambda = 0, 3, 15$

Eigen Vectors are $X_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ $X_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$ $X_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

The Eigen Value Decomposition is given by the formula

$$A = PDP^{-1}$$

For a symmetric matrix (where row elements are equal to column elements), we

have the formula $A = PDP^T$

where:

$P = \text{Initial Matrix}$

$D = \text{Diagonalized Eigen Values}$

$P^T = \text{Transposed Initial Matrix}$

From this we can derive the formula

$$A = \lambda_1 X_1 X_1^T + \lambda_2 X_2 X_2^T + \dots + \lambda_n X_n X_n^T$$

Representing the above problem in matrix form

$$A = PDP^T$$

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

The orthonormal for eigen vectors can be calculated as

$$X_1 = \sqrt{(1)^2 + (2)^2 + (2)^2} = \sqrt{9} = 3$$

$$X_2 = \sqrt{(2)^2 + (1)^2 + (-2)^2} = \sqrt{9} = 3$$

$$X_3 = \sqrt{(2)^2 + (-2)^2 + (1)^2} = \sqrt{9} = 3$$

$$X_1 = \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix} \quad X_2 = \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix} \quad X_3 = \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix}$$

$$A = 0 \begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix} \begin{bmatrix} 1/3 & 2/3 & 2/3 \end{bmatrix} + 3 \begin{bmatrix} 2/3 \\ 1/3 \\ -2/3 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 & -2/3 \end{bmatrix}$$

$$+ 15 \begin{bmatrix} 2/3 \\ -2/3 \\ 1/3 \end{bmatrix} \begin{bmatrix} 2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} = A$$